DP Optimizations

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1 Divide and Conquer Optimization

1.1 Problem

We are solving the following dp criteria:

$$dp(i,j) = \min_{i \le k \le j} (dp(i-1,k-1) + C(k,j)),$$

where $dp(i,j)$ denotes partitioning the first $j+1$ elements into $i+1$ subarrays and $C(l,r)$ is the cost of placing the subarray $a[l] \dots a[r]$ into one contiguous segment.

1.2 opt(i,j)

$opt(i,j)$ denotes the value of k achieving this minimum.

1.3 Sufficient condition for Divide and Conquer DP

$opt(i,j)$ needs to be monotonic, that is:

$$opt(i,j) \le opt(i,j+1)$$

1.4 Special Case

One sufficiency condition for opt to be monotonic, is for the cost function to satisfy Quadrangle inequality.

1.4.1 Quadrangle Inequality

for $a \le b \le c \le d$,

$$C(a,c) + C(b,d) \le C(a,d) + C(b,c)$$